Epithalon

Tetrapeptide Telomerase Activator | 10mg & 50mg x 10 Vials

Telomere Length | Anti-Aging | Pineal Gland Regulation | SubQ / IM

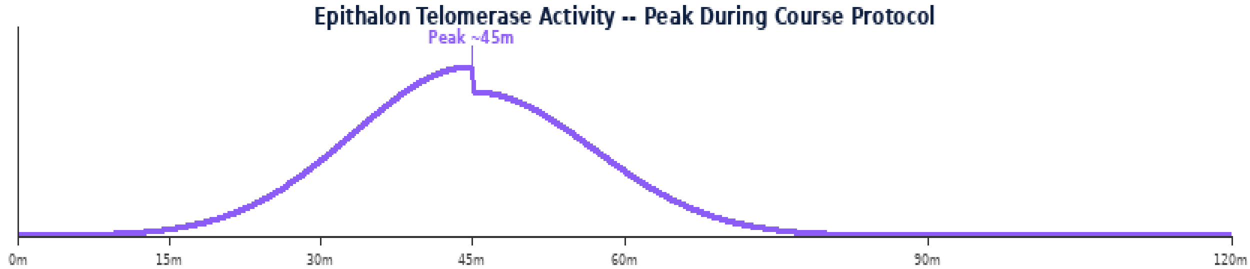
Purity: 99.8% | Endotoxin-Free | Research Use Only

Epithalon (Epithalamin) is a synthetic tetrapeptide (Ala-Glu-Asp-Gly) derived from the bovine pineal gland extract Epithalamin. It is the most researched peptide for telomere elongation and longevity, with Russian clinical research dating back to the 1980s. Epithalon activates telomerase, elongates telomeres, regulates the pineal gland, and has demonstrated life extension in multiple animal models.

Epithalon Vial Comparison

Parameter	Specification
Class	Synthetic tetrapeptide -- telomerase activator, pineal bioregulator
Vial Sizes	10mg x 10 | 50mg x 10 | Purity 99.8%
Reconstitution	1 mL BAC Water: 10mg->10mg/mL | 50mg->50mg/mL
Standard Dose	1,000-10,000 mcg (1-10mg) per injection SubQ or IM
Frequency	1-2x daily SubQ/IM | typically 10-day courses
Key Effects	Telomerase activation, telomere elongation, pineal regulation, antioxidant, life extension models
Cycle & Storage	10-20 day courses, 2-6 months between | 2-8 deg C | 28 days

MECHANISM OF ACTION & RESEARCH BENEFITS



Pathway	Mechanism	Benefit
Telomerase Activation	Induces expression of hTERT (telomerase reverse transcriptase); elongates shortened telomeres in somatic cells; delays cellular senescence; extends replicative lifespan of cells	Anti-aging; telomere biology; cellular senescence research
Pineal Regulation	Regulates melatonin synthesis and secretion from pineal gland; restores circadian rhythm regulation; normalises neuroendocrine function declined with age; antioxidant via melatonin pathway	Sleep regulation; neuroendocrine anti-aging; circadian research
Antioxidant & Immunity	Reduces lipid peroxidation; antioxidant enzyme upregulation; normalises T-cell immune function; reduces autoimmune markers in aged animals; life extension demonstrated in multiple species	Immune restoration; antioxidant; longevity research

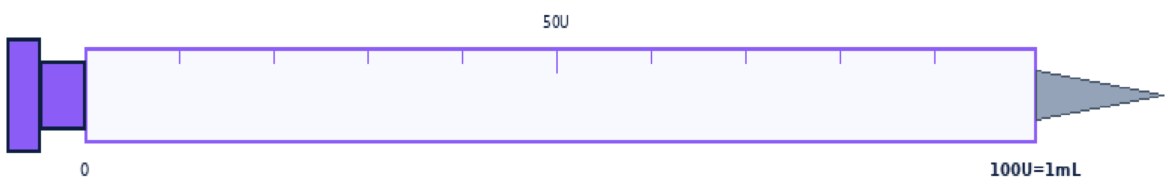
Benefit	Context
Telomere Elongation	Epithalon is the most studied peptide for telomerase activation and telomere length -- with multiple published studies demonstrating elongation.
Life Extension Research	Demonstrated life extension up to 24% in animal models with repeated course protocols. Longest-researched anti-aging peptide.
Pineal Bioregulator	Restores pineal gland function and melatonin synthesis that declines with age -- addressing a core mechanism of neuroendocrine aging.

RECONSTITUTION & DOSE CALCULATION

All vials use 1 mL BAC Water. Room temperature. Swirl gently -- never shake.

1 mL BAC Water per vial. Roll gently -- NEVER shake.

Step	Action	Detail
1	Gather	Vial, BAC Water, insulin syringe (29-31G), swabs, sharps.
2	Clean stoppers	Wipe both stoppers. Air-dry 10-15 sec.
3	Draw 1mL BAC	Exactly 1 mL. Expel air bubbles.
4	Inject into vial	Aim at GLASS WALL. Slowly. Never onto powder.
5	Swirl & label	Roll 20-30 sec. Clear within 60 sec. Never shake. Label date. 2-8 deg C. Use within 28 days.



1mL Insulin Syringe | 29-31G | SubQ

Formula: Vol(mL)=Dose(mcg)/Conc(mcg/mL) **| Units:** Vol x 100

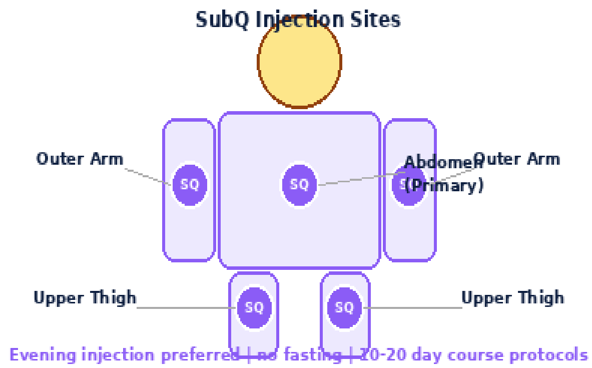
10mg x 10 -- 1mL BAC -> 10.0mg/mL (10,000mcg/mL)

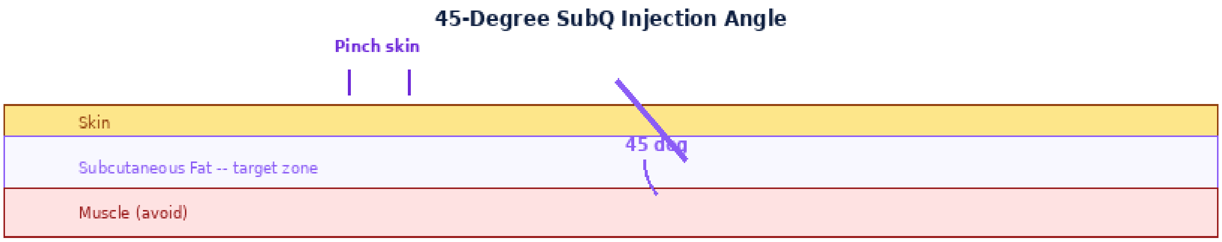
Dose(mcg)	1000mcg	3000mcg	5000mcg	10000mcg
Vol(mL)	0.1000	0.3000	0.5000	1.0000
Units	10.0U	30.0U	50.0U	100.0U
Doses/Vial	10	3	2	1

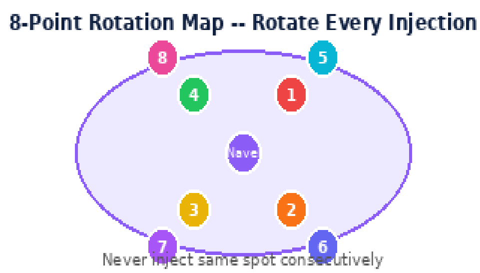
50mg x 10 -- 1mL BAC -> 50.0mg/mL (50,000mcg/mL)

Dose(mcg)	1000mcg	3000mcg	5000mcg	10000mcg
Vol(mL)	0.0200	0.0600	0.1000	0.2000
Units	2.0U	6.0U	10.0U	20.0U
Doses/Vial	50	16	10	5

INJECTION TECHNIQUE, PROTOCOLS & GOLDEN RULES







Step	Action	Detail
1	Wash & prepare	Wash hands. 29-31G SubQ or 23-25G IM. No fasting.
2	Draw dose	Standard 1,000-5,000 mcg. Draw exact volume.
3	Inject SubQ/IM	Abdomen SubQ or deltoid IM. Rotate sites.
4	Evening timing	Evening injection preferred -- aligns with melatonin/pineal cycle.

Protocol	Dose	Frequency	Duration	Notes
Standard Course	1,000-5,000 mcg	1-2x daily x 10-20 days	Course every 2-6 months	Evening injection preferred.
High-Dose Course	5,000-10,000 mcg	1x daily x 10 days	Course every 3-6 months	Research maximum dose.

Side Effect	Likelihood	Management
Injection site reaction	Common	Mild. Rotate sites.
Vivid dreams / sleep change	Occasional	Pineal/melatonin effect. Usually positive.
Fatigue	Rare	Cellular adaptation. Resolves.

Contraindications: Cancer history, pregnancy, under 18.

1. 10mg: 1mL BAC->10mg/mL | 50mg: 1mL->50mg/mL.
2. No fasting required. Evening injection preferred (pineal/circadian alignment).
3. Administer in 10-20 day courses with 2-6 months between courses.
4. Standard course: 1,000-5,000 mcg daily for 10-20 days.
5. Telomerase activation and telomere elongation are the primary research endpoints.
6. Rotate injection sites. 29-31G SubQ or 23-25G IM.
7. Store 2-8 deg C. Use within 28 days. Never freeze.
8. Most researched peptide for longevity -- extensively studied since 1980s.
9. Not for use during pregnancy, lactation, or in individuals under 18.